MIL-E-1/1405C
26 November 1974
SUPERSEDING
MIL-E-1/1405B
20 March 1973

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

TYPE 7640

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Forward-wave amplifier, frequency range 2.9 to 3.1 GHz, power output 1 kw, periodic permanent magnet focus, air cooled, coaxial rf connectors

ABSOLUTE RATINGS: See note 2

Parameter:	Ef	If(surge)	ek	ik	ia + iw	VSWR
Unit	V	A	v	a	ma	---
Maximum:	8.5	10	-8,000	2.0	1,000	2.0:1.0
Minimum:	6.0	---	---	---	---	---

ABSOLUTE RATINGS: See note 2

Parameter:	Pp	Du	tp	Pi(rf)	tk	TE
Unit:	W	---	μs (Note 3)	W (Note 9)	sec	°C (Note 4)
Maximum:	100	0.006	6.0	2	---	150
Minimum:	---	---	---	---	120	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	DC connectors:	See figure 1
Mounting position:	Any	Focusing circuit:	PPM, integral
Cooling:	2 cfm minimum air through base	Weight:	17.5 pounds, maximum
RF connectors:	See figure 1	Marking:	See notes 6 and 7

TEST CONDITIONS:

Parameter:	Ef	Du	ek	tp	Pi(rf)	F
Unit:	V	---	v	μs	W	GHz
Test condition 1:	6.3	0.0055	Adj	5.5	0.001	F1, F2, F3
Test condition 2:	6.3	0.0055	Adj	5.5	1.0	F1, F2, F3
Test condition 3:	6.3	0.0055	Adj	5.5	1.0	F2

Frequency		
F	GHz	±
1	2.9	2%
2	3.0	2%
3	3.1	2%

GENERAL:

Qualification - Not required

Burn-in - 25 hours (see note 17)

Service-life guarantee - See note 16

7640

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	5	-					
1301	Heater current	-	-		If	---	4.5	A
1256	Cathode current	-	2	Pi(rf) = 0	ik	0.75	1.75	a
4303	Cathode warmup time	-	2	tk1 = 250 sec; tk2 = 300 sec; Pi(rf) = 0	Δik	---	0.1	a
1261	Cathode voltage	-	1		ek	-6.0	-7.0	kv
1261	Cathode voltage	-	2		ek	-6.5	-7.5	kv
4253	Small-signal gain (1)	1, 9	1	σ' = 1.5 (max)	Gss	33	43	dB
4253	Small-signal gain (2)	1, 9	1	σ' = 1.5 (max)	Gss	30	---	dB
4253	Small-signal gain variation	1, 9, 10	1	σ' = 1.5 (max)	ΔGss	---	3.0	dB
4253	Large-signal gain (1)	1, 9	2	σ' = 1.5 (max)	Gls	30	40	dB
4253	Large-signal gain (2)	1, 9	2	σ' = 1.5 (max)	Gls	27	---	dB
4253	Large-signal gain variation (1)	1, 9	2	σ' = 1.5 (max)	ΔGls	---	2.0	dB
4253	Large-signal gain variation (2)	1, 9, 14	3	σ' = 1.5 (max)	ΔGls	---	2.5	dB
4251	Power output variation	8, 9, 12	2	σ' = 1.5 (max); F = F1; ek (1) = 1.02 ek; Pi(rf) (1) = 0.5 W; Pi(rf) (2) = 1.5 W; F = F3; ek(2) = 0.98 ek	Δpo	---	1.0	dB
4261	Spurious power output (1)	13	1	Pi(rf) = 0	spo	---	-10	dBm
4261	Spurious power output (2)	13	2	Pi(rf) = 0	spo	---	-10	dBm
	Quality conformance inspection, part 2							
---	Spectrum	9, 15	1	F = F2	ΔF	---	30	MHz
4253	Large-signal gain variation (3)	9, 11	3	tk3 = 5 minutes; tk4 = 20 minutes	ΔGls	---	1.5	dB

NOTES:

1. A swept frequency from F1 through F3 may be used for this test.
2. Cathode and one leg of heater shall be connected internally. Anode, helix, and collector shall be at ground potential. All voltages are referred to ground except heater potential, which is referred to cathode.
3. The pulse shall comply with the requirements stated in method 4304, pulse characteristics, except that the rise and decay times shall not exceed 0.5 and 0.6 μs respectively. The peak-to-peak ripple shall not exceed 100 volts.
4. TE is the temperature at the midpoint of the cylindrical surface of the bulb with the specified flow of cooling air.

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NOTES: -Continued

5. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

6. Each tube shall be marked with the following information:

   (a) Manufacturer's identification.
   (b) JAN 7640.
   (c) Serial number.
   (d) Acceptance date.
   (e) Cathode voltage for optimum small-signal gain.
   (f) Cathode voltage for optimum large-signal gain.
   (g) RF input and output connectors.

7. One reproducible and two prints of data sheets for all tests performed under this specification shall be supplied to the contractor. One copy of the data sheet shall be packed with the tube.

8. The rf output pulse shall evidence no more than 10 percent amplitude distortion from any source other than the applied dc pulse.

9. The input power, Pi(rf), is defined as incident power from a 50-ohm source at the rf input connector of the tube.

10. ΔGss and ΔGls(1) shall be the difference between maximum gain and minimum gain within the frequency range.

11. ΔGls(2) shall be the difference in Gls at tk3 and tk4.

12. Δpo shall be the difference in power output for Pi(rf) (1) and Pi(rf) (2).

13. The input shall be terminated into a 50-ohm matched load. The output termination shall be a 3-dB attenuator in series with a variable-phase short circuit. The bandpass filter shall have a bandpass from 2.1 to 3.9 GHz.

14. A variation of ±5 percent in ek from the test condition 2 value shall result in a gain variation, ΔGls, from the measured value, Gls, of not greater than the value specified.

15. Spectrum width shall be measured with a receiver having an IF bandwidth of 3.0 MHz, or less. In the frequency range 2.8 to 3.2 GHz, the range shall be the respective lowest and highest frequencies at which the signal power is at least 30 dB below the power of the center frequency.

16. Service-life guarantee of 1,000 hours is applicable when specified in the contract.

17. After completion of this test, all tubes shall meet the parameters listed on the data sheet (see note 7).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 80
DSA - ES

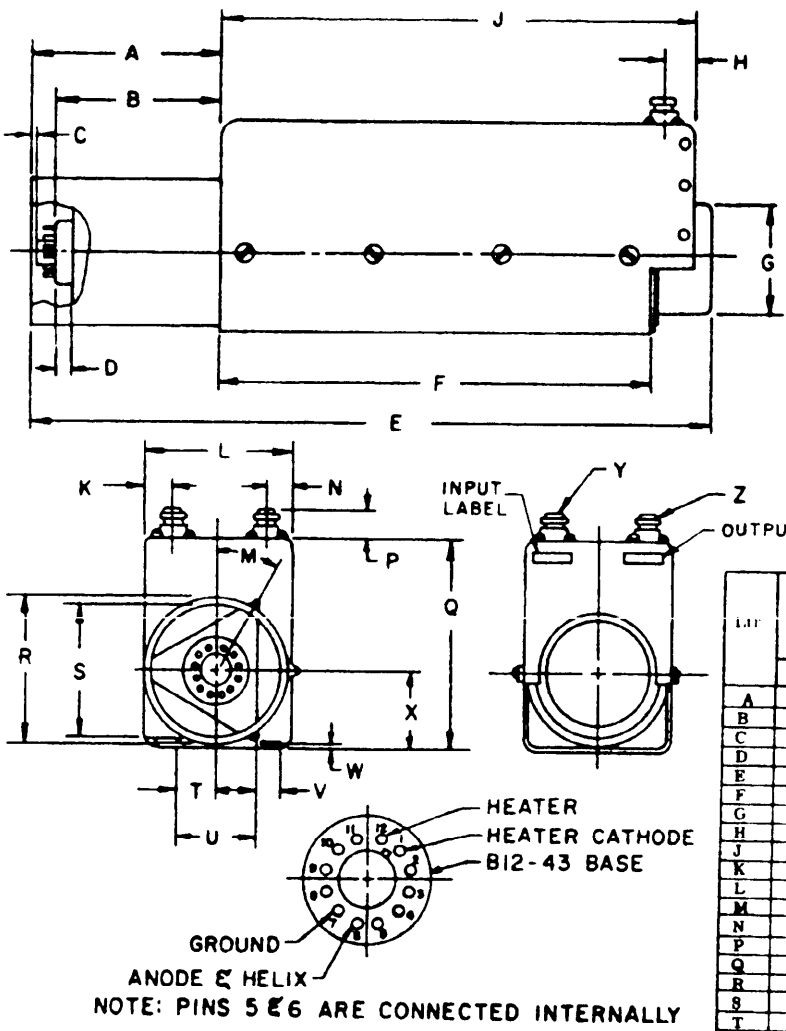
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2876)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
A	4.5625 (115.69)	4.6875 (119.06)
B	3.9687 (100.80)	4.0313 (102.40)
C	.0937 (2.38)	.1563 (3.97)
D	.375 (9.53)	.500 (12.70)
E		16.625 (422.28)
F	10.4687 (265.90)	10.5313 (267.50)
G	2.657 (67.49)	2.719 (69.06)
H	.7187 (18.25)	.7813 (19.85)
J	11.4687 (291.30)	11.5313 (292.90)
K	.7187 (18.25)	.7813 (19.85)
L	3.5679 (90.62)	3.6405 (92.47)
M	30° NOM	
N	.7187 (18.25)	.7813 (19.85)
P	.5937 (15.08)	.6563 (16.67)
Q	4.9687 (126.20)	5.0313 (127.80)
R	3.437 (87.30)	3.499 (88.87)
S	3.219 (81.76)	3.281 (83.34)
T	.9314 (23.66)	.9436 (23.97)
U	1.814 (46.06)	1.936 (49.17)
V	.5937 (15.08)	.6563 (16.67)
W	.059 (1.50)	.069 (1.75)
X	1.788 (45.42)	1.820 (46.23)
Y	INPUT UG - 19 B/U connector	
Z	OUTPUT UG-19B/U connector	

FIGURE 1. Outline drawing of electron tube type 7640.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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